Project Course Presentation

ME 299

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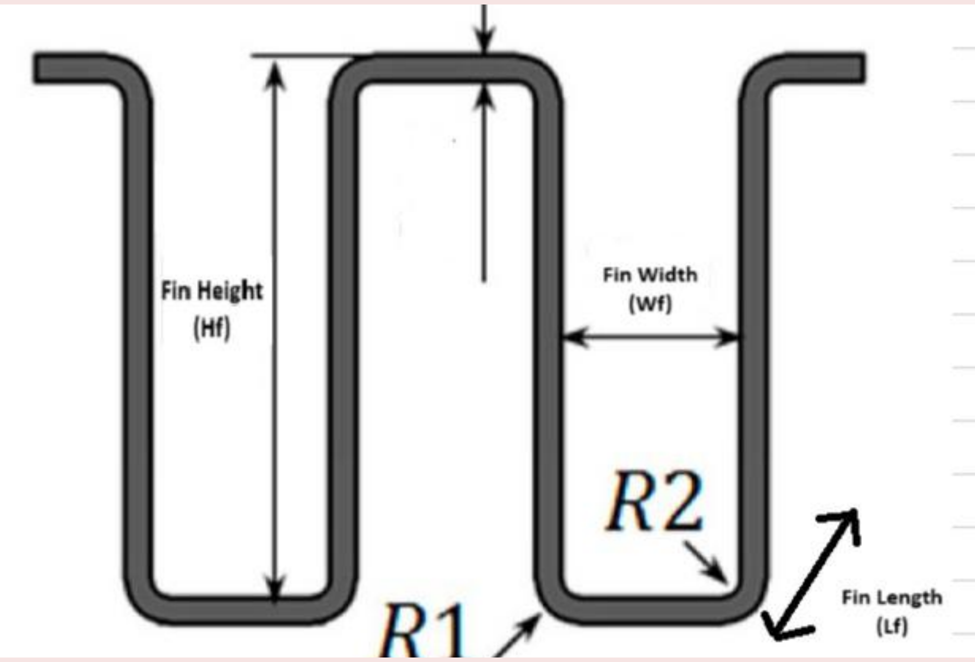
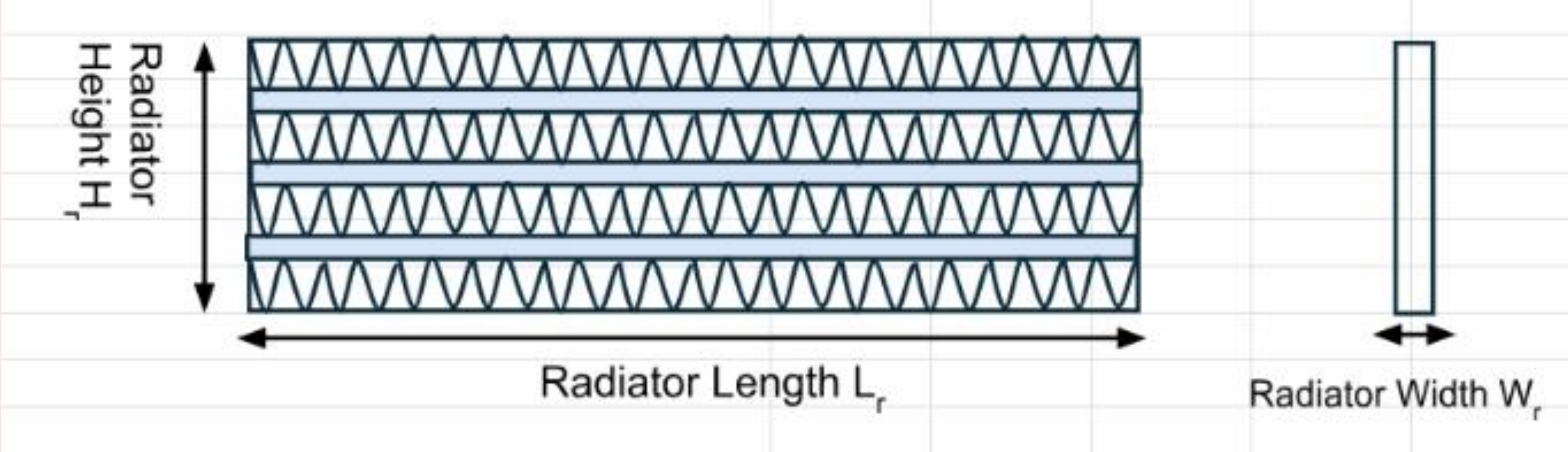
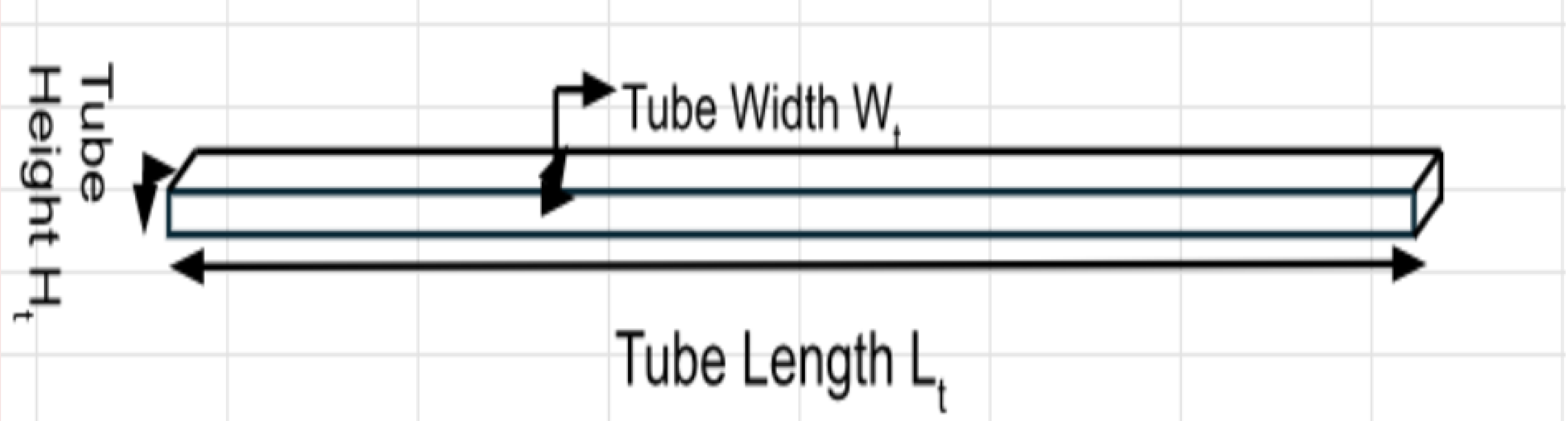
Radiator Performance Enhancement in System







[Radiator]

Radiator Performance Enhancement in System

- The radiator is a key part of any thermal management system, responsible for carrying away excess heat and keeping the overall setup within safe operating temperatures.
- In many cases, radiators do not perform at their highest possible efficiency because of uneven airflow, limited heat-transfer area, or restrictions in coolant flow.
- Improving the radiator's efficiency directly helps the system run more reliably, and maintain better performance over time.
- In our project, the focus was on improving the radiator's performance through design-based changes, rather than adding more power or complexity.

What we want to achieve

- Improve the heat-rejection capability of the radiator without increasing power consumption or system complexity.

- Identify how airflow, fin geometry, and coolant flow influence the overall cooling performance.

- Find a radiator–fan combination that provides stable cooling even when ambient temperature changes

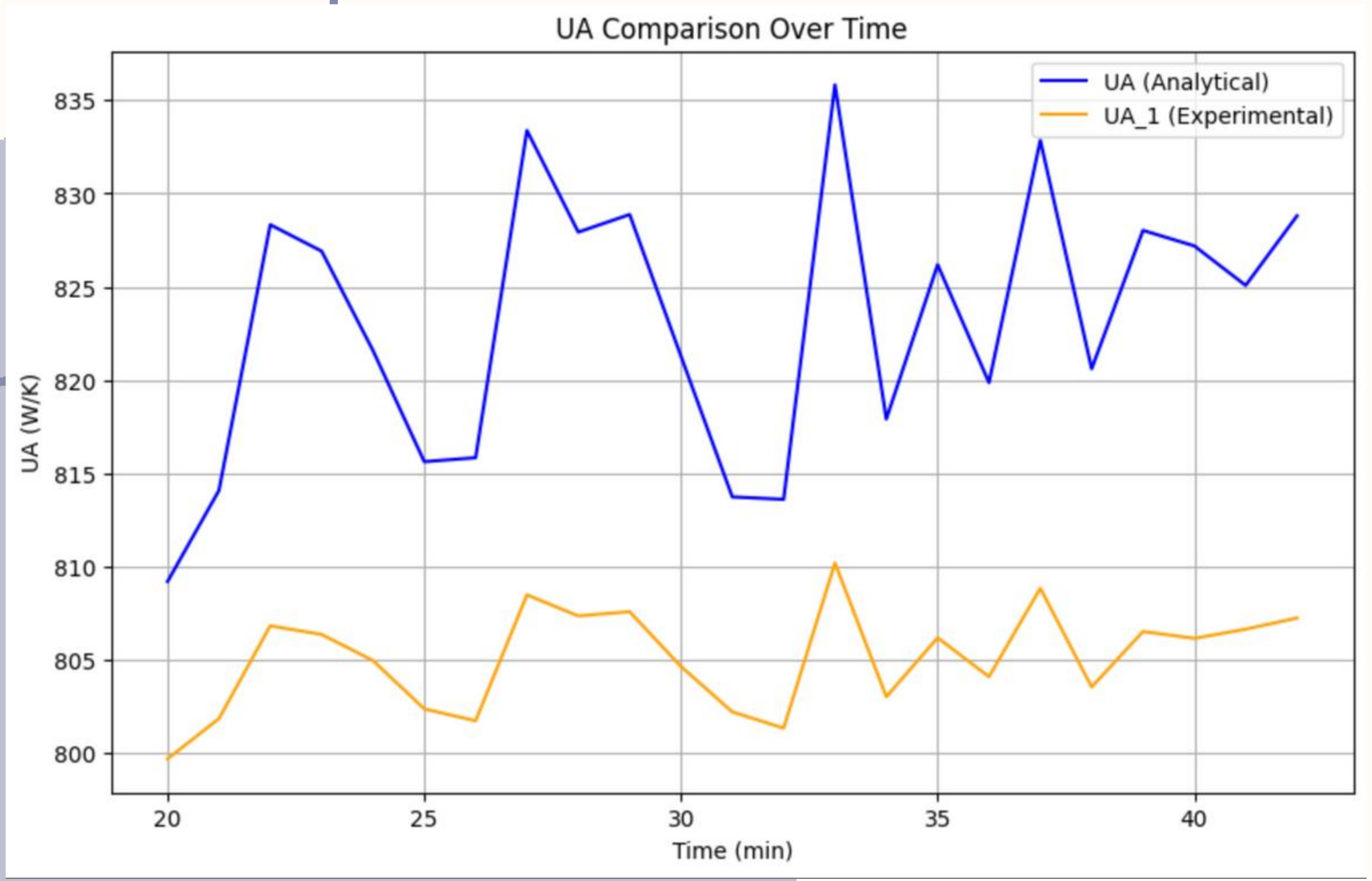
Work we did so far

- Calculated radiator geometry parameters: tube dimensions, fin height/width, fin count, and total air-side and coolant-side surface areas.
- Evaluated different radiator configurations and compared how each design affects available heat-transfer area.
- Plotted outlet air temperature for different inlet air velocities to study how airflow influences cooling.
- Generated Q vs. ambient temperature plots to understand radiator behaviour under varying environmental conditions.
- Estimated coolant-side heat removal using measured inlet–outlet temperatures and flow-rate data.

Work we did so far

- Studied the heat-exchangers chapter from Incropera to understand the theoretical background.
- Used the LMTD method to determine the effective temperature difference across the radiator core.
- Calculated the overall heat-transfer coefficient by combining air-side and coolant-side contributions.
- Matched air-side temperature rise with coolant temperature drop to check the consistency of the model.
- Reviewed multiple radiator and fan options available commercially and compared their dimensions and airflow ratings.
- Shortlisted viable radiator–fan combinations based on size, airflow, and estimated cooling performance.
- Plotted the overall heat transfer coefficient (U) to compare analytical (a formula from some research paper) and experimental result.

Observations



$$UA = \frac{1000\, C_{\min}}{\pi\,(C_r^{0.15}+\varepsilon)} \left[\max\left(\left(\frac{1+0.44(1-C_r)}{1-\eta_1+0.44(1-C_r)+\varepsilon} \right)^{2.5} - 0.92,\ 0 \right) \right]^{0.8}$$

Difference between analytical and experimental values of UA(overall heat transfer coefficient times area)

UA_1=Q_dot/LMTD

Observation
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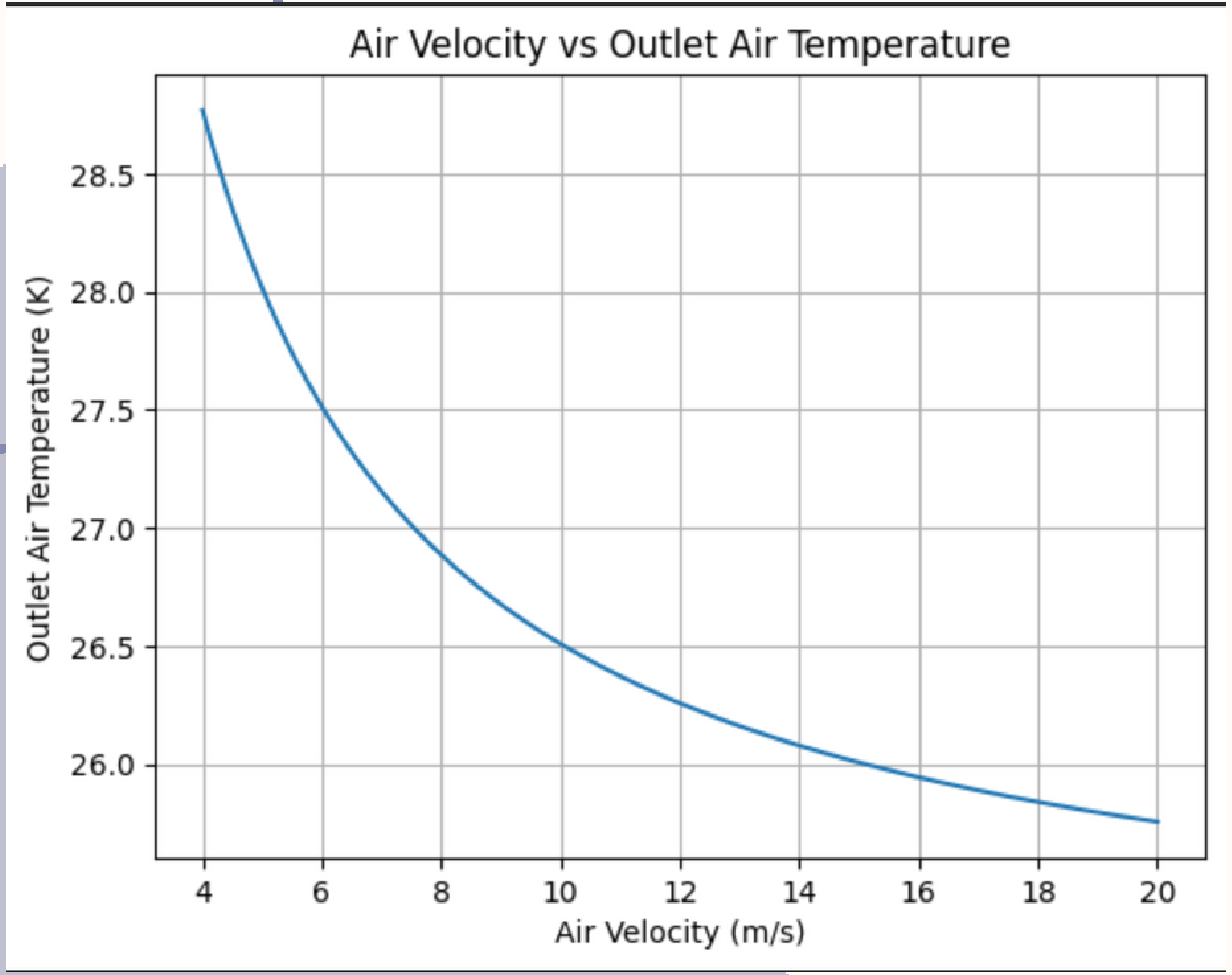

Air Velocity vs Outlet Air Temperature

Here, we assumed mass flow rate, inlet and outlet temperatures for water and for a given air velocity at a given air inlet found the outlet temperature.

If the outlet temperature reduces we can also say for that inlet temperature and velocity for air if conditions of water flow change the Power output for radiator can be improved.

So we found out that velocity of air in the range of 8 to 12 m/s can give big Power outputs and also practical to be kept in real systems.

Observation
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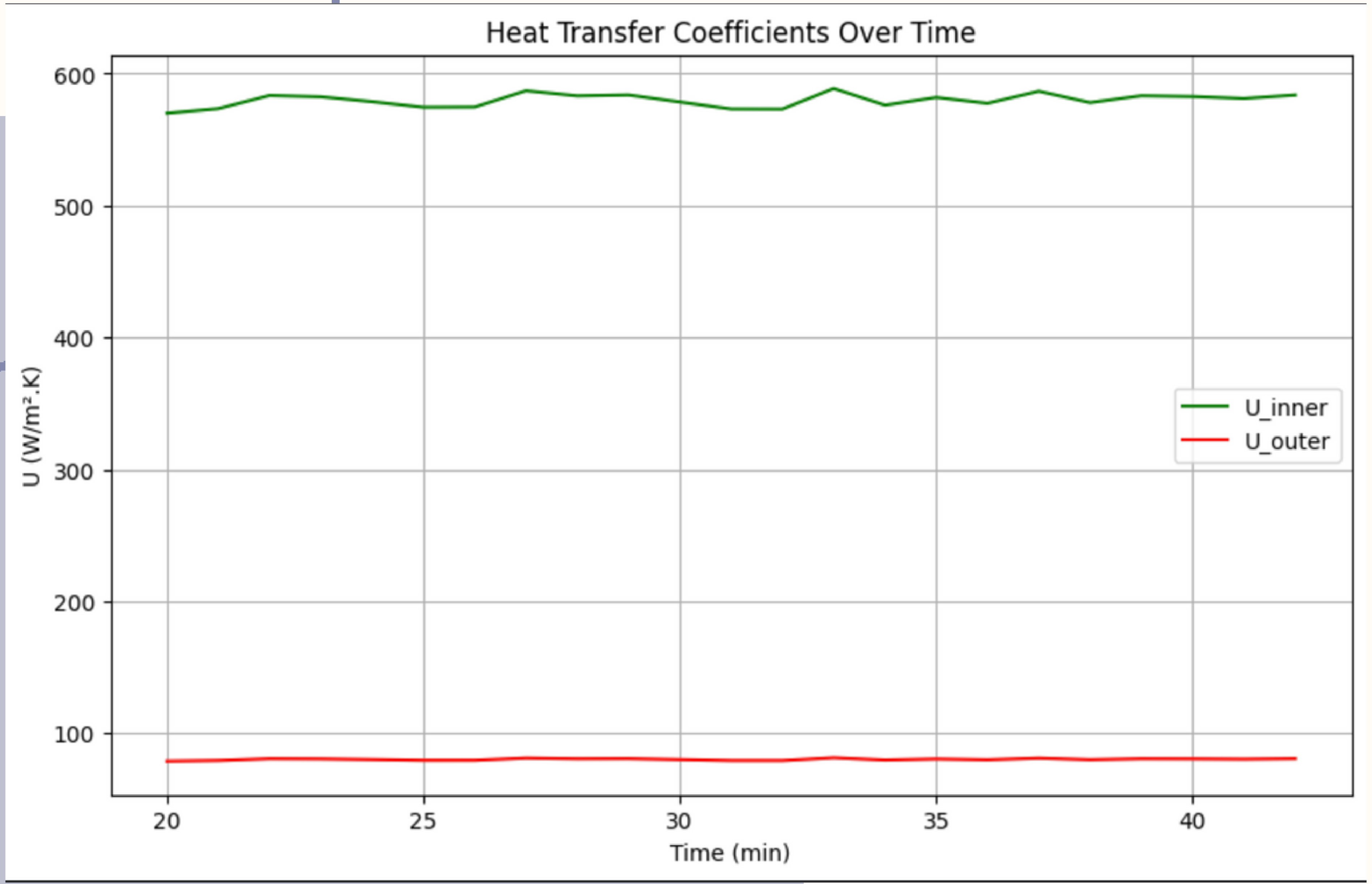
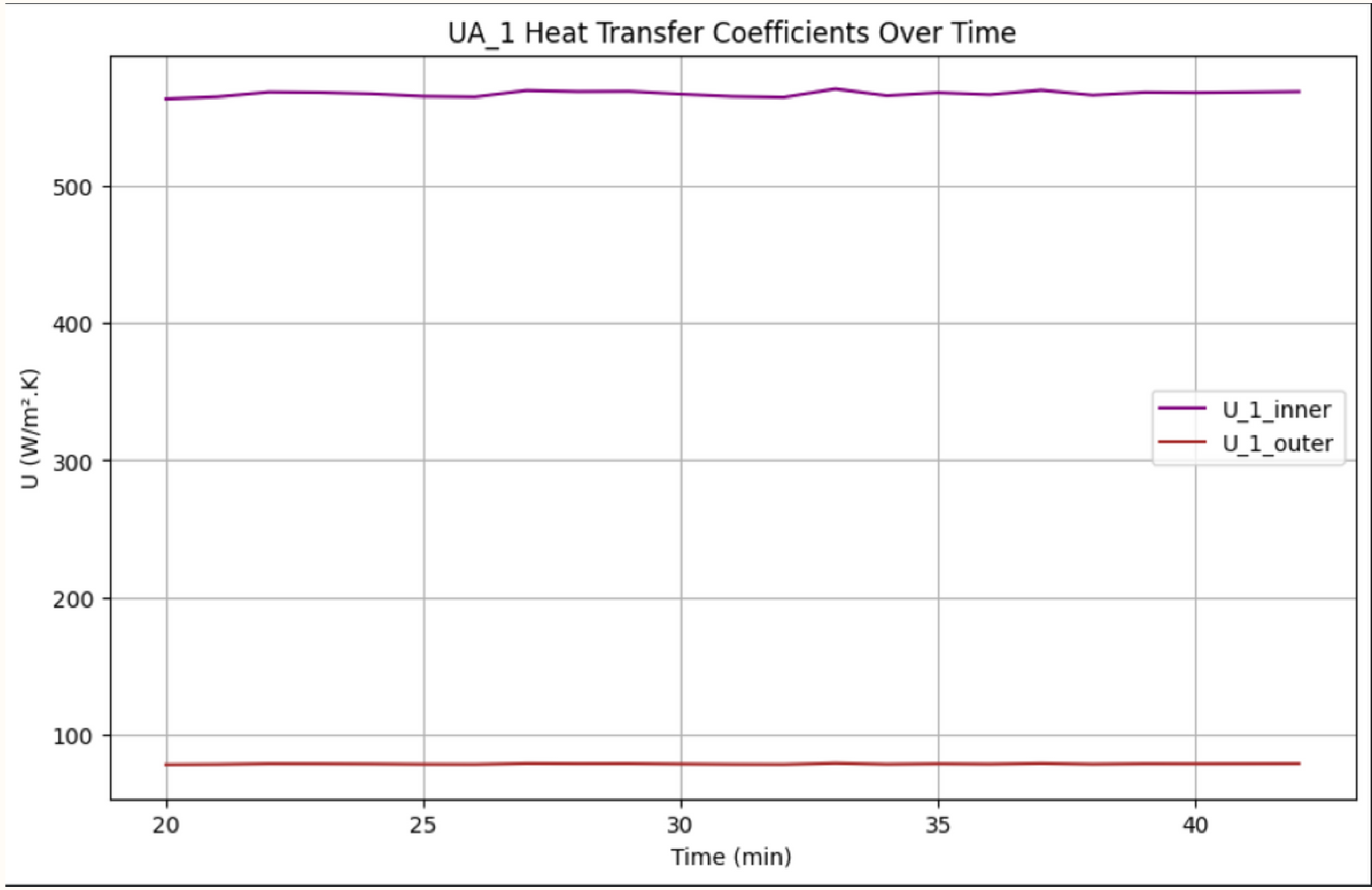


Plot of Heat Transfer Coefficients of Fluid flow(inner) and Air flow(outer) found Analytically



Plot of Heat Transfer Coefficients of Fluid flow(inner) and Air flow(outer) found Experimentally

Observation
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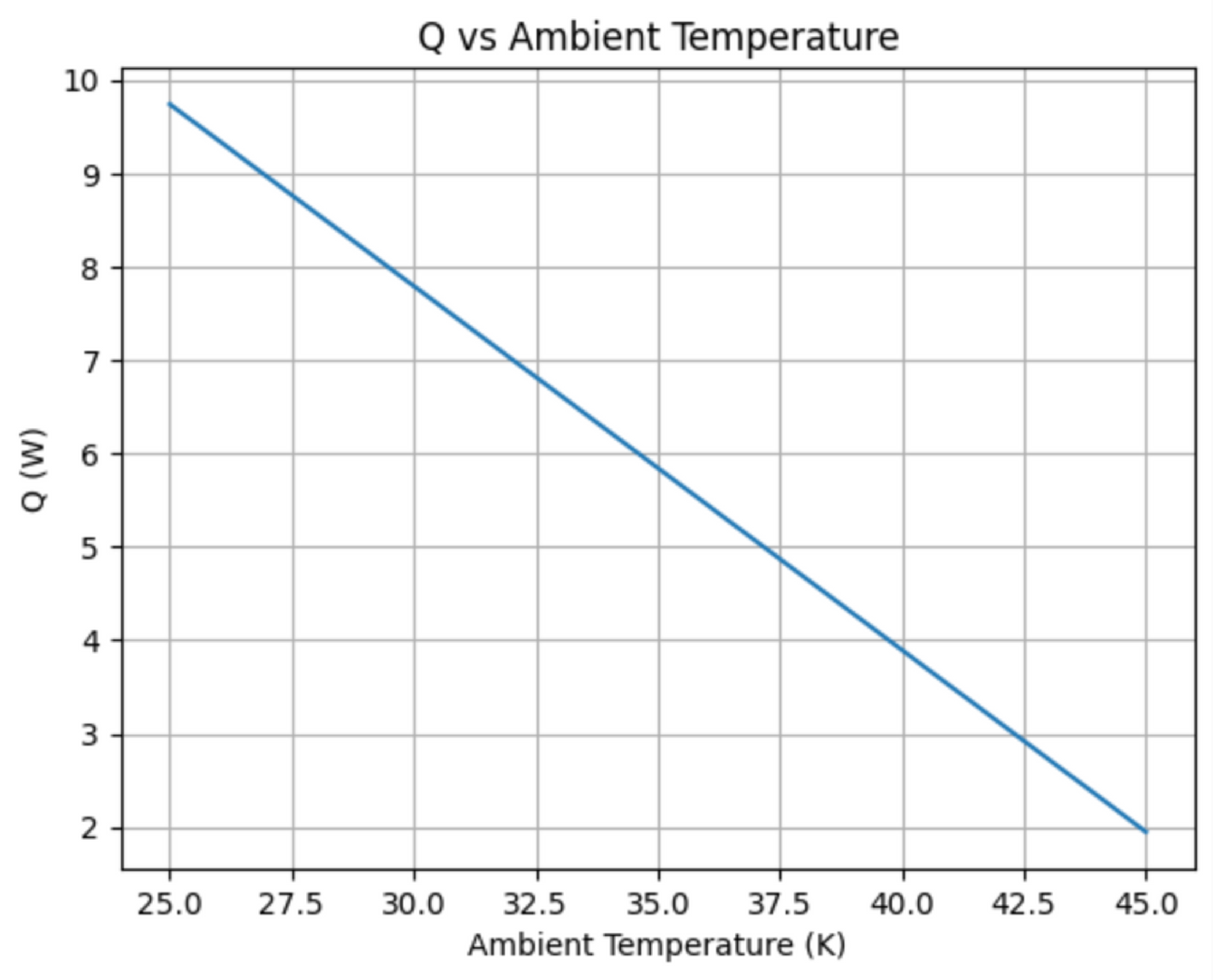

Power output of radiator vs Ambient temperature found out using Q=mcΔT

Here we tried to vary the ambient temperature to find how it affects the power output of radiator at some constant output temperature of air and constant water conditions.

Observation
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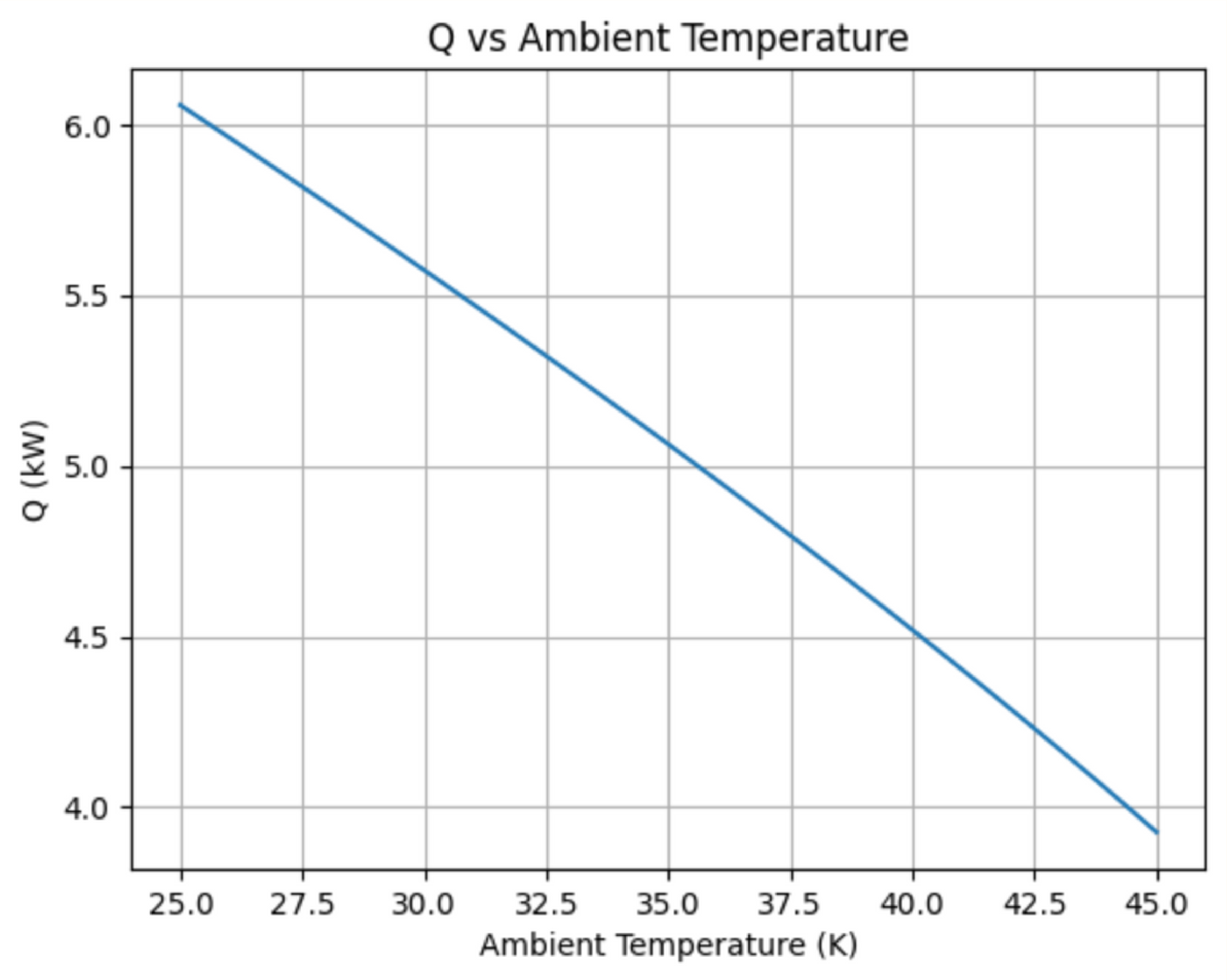

Power output of radiator vs Ambient temperature found out using Q=UA(LMTD)

Conclusions

- There was some deviations between analytical and experimental values of Heat Transfer Coefficient.
- On increasing the speed of cooling fan velocity of air passing through the radiator increases resulting in more Power Output of the Radiator.
- Optimal range for fan speed is between 8 to 12 m/s.
- Radiator Power Output increases if the ambient temperature is reduced.

Project Work

Onedrive Link

This link contains:

1)The python file we used to do relevant calculations and plotting the graphs.

2)The excel sheet that contains the area calculations for different radiators.

3)Power_15 file has all the data used for heat transfer coefficient calculation.

Acknowledgements

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Thank you!